

SRI CHAITANYA IIT ACADEMY, INDIA

AP • TELANGANA • KARNATAKA • TAMILNADU • MAHARASHTRA • DELHI • RANCHI

PRAGYA G8 – ACHIEVERS | MATHEMATICS
QUADRATIC EQUATIONS | LEVEL 3*Tricky – the method has to be found before it can be used.*

1. If α, β are the roots of $x^2 + px + q = 0$, then the equation whose roots are $(\alpha - \beta)^2$ and $(\alpha + \beta)^2$ is
- (A) $x^2 + (2p^2 - 4q)x + p^2(p^2 - 4q) = 0$ (B) $x^2 - (2p^2 - 4q)x + p^2(p^2 - 4q) = 0$
- (C) $x^2 - (p^2 - 4q)x + p^2 = 0$ (D) $x^2 - (2p^2 + 4q)x - p^2(p^2 - 4q) = 0$
2. The equation whose roots are the square of the sum and the square of the difference of the roots of $2x^2 + 2(m + n)x + m^2 + n^2 = 0$ is
- (A) $x^2 - 4mnx - (m^2 - n^2)^2 = 0$ (B) $x^2 - 2mnx - (m^2 + n^2)^2 = 0$
- (C) $x^2 - 4mnx + (m^2 - n^2)^2 = 0$ (D) $x^2 + 4mnx + (m^2 - n^2)^2 = 0$
3. The sum of the values of x satisfying $2\sqrt{x} + 2x^{-1/2} = 5$ is
- (A) 4 (B) $\frac{17}{4}$
- (C) $\frac{5}{4}$ (D) $\frac{15}{4}$
4. The sum of the values of x satisfying $2^{2x+3} - 57 = 65(2^x - 1)$ is
- (A) 0 (B) 3
- (C) 6 (D) -3
5. The values of the parameter a for which the quadratic equations $(1 - 2a)x^2 - 6ax - 1 = 0$ and $ax^2 - x + 1 = 0$ have at least one root in common are
- (A) $\frac{2}{9}$ (B) $0, \frac{1}{2}$
- (C) $\frac{1}{3}, \frac{1}{2}, \frac{2}{9}$ (D) $\frac{1}{2}, \frac{2}{9}$
6. The curve $y = ax^2 + bx + c$ lies entirely above the x -axis with its vertex in the second quadrant, and α, β are the roots of $ax^2 + bx + c = 0$ (D is the discriminant). Then
- (A) $a > 0, b < 0, c > 0, D < 0, \alpha + \beta > 0, \alpha\beta > 0$
- (B) $a > 0, b > 0, c > 0, D < 0, \alpha + \beta < 0, \alpha\beta < 0$
- (C) $a > 0, b > 0, c > 0, D < 0, \alpha + \beta < 0, \alpha\beta > 0$
- (D) $a > 0, b > 0, c > 0, D > 0, \alpha + \beta > 0, \alpha\beta > 0$

7. If p is a positive odd integer, the roots of $x^2 - px + q = 0$ are prime numbers and $p + q = 23$, then the absolute value of the difference of the roots is
(A) 3 (B) 5
(C) 2 (D) 1
8. If α, β are the roots of $x^2 - p(x + 1) - c = 0$, $c \neq 1$, then $(\alpha + 1)(\beta + 1)$ equals
(A) $1 + c$ (B) $1 - c$
(C) $-c$ (D) $c - 1$
9. The set of values of k for which $\left| \frac{x^2 + kx + 1}{x^2 + x + 1} \right| < 2$ for every real x is
(A) $-8 < k < 4$ (B) $k > 0$
(C) $-1 < k < 7$ (D) $0 < k < 4$
10. If x and y are real numbers connected by $9x^2 + 2xy + y^2 - 92x - 20y + 244 = 0$, then the range of x is
(A) $[1, 10]$ (B) $[2, 6]$
(C) $[3, 10]$ (D) $[3, 6]$
11. If x is real, then the range of $\frac{x^2 + 14x + 9}{x^2 + 2x + 3}$ is
(A) $[-4, 5]$ (B) $[-4, 4]$
(C) $[-5, 5]$ (D) $[-5, 4]$
12. The smallest and greatest values of $\frac{x^2 + x + 1}{x^2 + 1}$ for $x \in \mathbb{R}$ are
(A) 0 and 2 (B) $\frac{1}{3}$ and 3
(C) $\frac{1}{2}$ and $\frac{3}{2}$ (D) 1 and $\frac{3}{2}$
13. If α, β are the roots of $ax^2 + bx + c = 0$, then the equation whose roots are $\alpha + \frac{c}{a}$ and $\beta + \frac{c}{a}$ is
(A) $a^2x^2 - (ca + b)x + c(a + b) = 0$ (B) $a^2x^2 + 2(ac + b)x - c(a + b + c) = 0$
(C) $a^2x^2 - a(-b + 2c)x + c(a - b + c) = 0$ (D) $a^2x^2 - 2(ac + b)x + c(a + b) = 0$
14. If $y = \frac{x^2 + 2x + c}{x^2 + 4x + 3c}$ takes all real values, then
(A) $c > 1$ (B) $c > 0$
(C) $c < -1$ (D) $0 < c < 1$
15. If $\frac{x^2 + ax + 1}{x^2 + x + 1} < 3$ for all real x , then

(A) $-1 < a < 7$

(B) $a < -1$

(C) $a < 0$

(D) $a > 7$